



Applications of linear superposition principle to resonant solitons and complexitons



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ABSTRACT

We apply the linear superposition principle to Hirota bilinear equations and generalized bilinear equations. By extending the linear superposition principle to complex field, we construct complex exponential wave function solutions first and then get complexions by taking pairs of conjugate parameters. A few examples of mixed resonant solitons and complexitons to Hirota and generalized bilinear differential equations are presented.

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1. Introduction

It is always of great interest to find exact solutions to nonlinear partial differential equations especially soliton equations. The Hirota direct method [1,2] is a powerful tool of finding exact soliton solutions to nonlinear partial differential equations which take advantage of bilinear derivatives. There are also some other works successfully extending the direct method [3–6]. For some kind of soliton equations like the Korteweg–de Vries equation [7] (KdV equation for short), Hirota established nonlinear superposition principles to find multi-soliton solutions.

It is well known that the KdV equation is a mathematical model of shallow water waves on surfaces. The Kadomtsev–Petviashvili equation [8] (KP equation for short) is a generalization of the KdV equation. Both the KdV equation and the KP equation are notable as two prototypical examples of exactly solvable models.

The linear superposition principle is very important in studying differential equations. However solutions to a nonlinear differential equation do not form a linear space. It is valuable that we can find some subset of solutions to a nonlinear differential equation which forms a linear space. In the Refs. [9,10], the authors discussed the linear superposition principle of exponential traveling waves. Recently the papers [11,12] extended the results to a special hyperbolic function ($\cosh(\eta)$) traveling waves and a trigonometric function ($\cos(\eta)$) traveling waves under certain conditions. Complexiton solutions, which are combinations of trigonometric function traveling waves and exponential traveling waves, are introduced by W. X. Ma in 2002 [13]. The Wronskian technique is an effective method to find complexitons [14,15]. However, the complexitons obtained by the Wronskian technique are very complicated for high dimension. In this paper, we establish the linear superposition principle for complexiton solutions which is generalization of [11,12].

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The paper is organized as follows. In Section 2, we study the linear superposition principle for generalized bilinear equations. In Section 3, we present a few illustrative examples, we get mixed resonant multi-soliton and multi-complexiton solutions, the previous results [4] are special cases of our result. Section 4 consists of conclusions and remarks.

2. Linear superposition principle for Hirota bilinear equations

2.1. Hirota derivatives

Suppose an integer $M \geq 1$, $x \in \mathbb{R}^M$. Let $D = (D_1, D_2, \dots, D_M)$, where D_j , $1 \leq j \leq M$, are Hirota bilinear derivatives with respect to x_j , $1 \leq j \leq M$, which are defined as follows.

Let f, g be differentiable functions in $\mathbb{R}^M$, $x = (x_1, x_2, \dots, x_M)$ and $x' = (x'_1, x'_2, \dots, x'_M)$. Then

$$D_j f \cdot g = D_{x_j} f \cdot g := (\partial_{x_j} - \partial_{x'_j}) f(x) g(x') \big|_{x'=x} = f_{x_j} g - f g_{x_j}. \quad (1)$$

The higher order Hirota derivatives are defined by

$$D_1^{n_1} D_2^{n_2} \cdots D_M^{n_M} f \cdot g = \prod_{m=1}^M (\partial_{x_m} - \partial_{x'_m})^{n_m} f(x) g(x') \big|_{x'=x}, \quad (2)$$

where $n_1, n_2, \dots, n_M$ are non-negative integers.

One of the important properties of Hirota derivatives is that when interchanging the functions f and g , we have

$$D_{j_1} D_{j_2} \cdots D_{j_k} f \cdot g = (-1)^k D_{j_1} D_{j_2} \cdots D_{j_k} g \cdot f, \quad (3)$$

by which it is easy to see that if k is an odd number, then

$$D_{j_1} D_{j_2} \cdots D_{j_k} f \cdot f = 0. \quad (4)$$

Let us consider the KdV equation

$$u_t + 6uu_x + u_{xxx} = 0. \quad (5)$$

We can use the transformation $u = 2(\log f)_{xx}$ to get the bilinear KdV equation

$$(D_x D_t + D_x^4) f \cdot f = 0. \quad (6)$$

The KP equation

$$(u_t + 6uu_x + u_{xxx})_x + u_{yy} = 0 \quad (7)$$

can be expressed as

$$(D_x D_t + D_x^4 + D_y^2) f \cdot f = 0 \quad (8)$$

under the transformation $u = 2(\log f)_{xx}$.

2.2. Bilinear equations and exponential wave solutions

Suppose that P is a polynomial in M variables with the properties that $P(0) = 0$ and $P(-x) = P(x)$. Let

$$\eta_m := \beta_{m,0} + \beta_{m,1}x_1 + \beta_{m,2}x_2 + \cdots + \beta_{m,M}x_M, \quad m = 1, 2, \quad (9)$$

where $\beta_{m,j}$, $m = 1, 2, j = 0, 1, \dots, M$ are constants. Then it is not difficult to prove that for any positive integer k , we have

$$D_j^k e^{\eta_1} \cdot e^{\eta_2} = (\beta_{1,j} - \beta_{2,j})^k e^{\eta_1 + \eta_2}. \quad (10)$$

Therefore

$$P(D) e^{\eta_1} \cdot e^{\eta_2} = P(\beta_{1,1} - \beta_{2,1}, \dots, \beta_{1,M} - \beta_{2,M}) e^{\eta_1 + \eta_2}. \quad (11)$$

In particular, we have

$$P(D) e^{\eta_1} \cdot e^{\eta_1} = P(0, \dots, 0) e^{2\eta_1} = 0, \quad (12)$$

since $P(0) = 0$. Hence function $f_1 = e^{\eta_1}$ with arbitrary constant coefficients $\beta_{1,j}$, $0 \leq j \leq M$, is a solution to the bilinear differential equation

$$P(D) f \cdot f = 0. \quad (13)$$

We now know that $f_1 = e^{\eta_1}$ and $f_2 = e^{\eta_2}$ are all solutions of (13). It is an interesting question whether a linear combination of f_1 and f_2 is still a solution or not. We are going to answer the question later.

2.3. Generalized Hirota bilinear derivatives

Let $p \in \mathbb{N}$ be given. For $f, g \in C^\infty(\mathbb{R}^M)$, the generalized Hirota bilinear derivatives are defined as follows [4]:

$$D_{p,1}^{n_1} D_{p,2}^{n_2} \cdots D_{p,M}^{n_M} f \cdot g := \prod_{m=1}^M (\partial_{x_m} + \alpha \partial_{x'_m})^{n_m} f(x) g(x')|_{x'=x}, \quad n_m \geq 0, \quad 1 \leq m \leq M, \quad (14)$$

where $x' = (x'_1, x'_2, \dots, x'_M)^T$, $n_1, \dots, n_M$ are nonnegative integers, and for any integer m , the m th power of α is defined by

$$\alpha^m := (-1)^{r(m)}, \quad (15)$$

where $r(m) \equiv m \pmod{p}$ with $0 \leq r(m) < p$.

When $p = 2k$ ($k \in \mathbb{N}$), all the general bilinear derivatives become Hirota derivatives, since $m - r(m)$ is even, we have $\alpha^m = (-1)^{r(m)} = (-1)^m$. Therefore $D_{2k,j} = D_j$ for $j = 1, \dots, M$.

Now we consider $p = 3$, we have

$$\alpha = -1, \quad \alpha^2 = \alpha^3 = 1, \quad \alpha^4 = -1, \quad \alpha^5 = \alpha^6 = 1, \dots \quad (16)$$

It is easy to get

$$\begin{aligned} D_{3,x} f \cdot g &= D_x f \cdot g = f_x g - f g_x, \\ D_{3,x} D_{3,y} f \cdot g &= D_x D_y f \cdot g = f_{xy} g - f_y g_x - f_x g_y + f g_{xy}, \\ D_{3,x} D_{3,y} D_{3,z} f \cdot g &= f_{xyz} g - f_{xy} g_z - f_{xz} g_y + f_x g_{yz} - f_{yz} g_x + f_y g_{xz} + f_z g_{xy} + f g_{xyz}, \\ D_{3,x} D_{3,y} D_{3,z} D_{3,t} f \cdot g &= f_{xyzt} g - f_{xyz} g_t - f_{xyt} g_z + f_{xy} g_{zt} - f_{xzt} g_y + f_{xz} g_{yt} + f_{xt} g_{yz} \\ &\quad + f_x g_{yzt} - f_{yz} g_x + f_{yz} g_{xt} + f_{yt} g_{xz} + f_y g_{xzt} + f_{zt} g_{xy} + f_z g_{xyt} + f_t g_{xyz} - f g_{xyzt}. \end{aligned}$$

Therefore, when $g = f$ we have

$$\begin{aligned} D_{3,x} f \cdot f &= f_x f - f f_x = 0, \\ D_{3,x} D_{3,y} f \cdot f &= 2(f_{xy} f - f_x f_y), \\ D_{3,x} D_{3,y} D_{3,z} f \cdot f &= 2f_{xyz} f, \\ D_{3,x} D_{3,y} D_{3,z} D_{3,t} f \cdot f &= 2(f_{ij} f_{kl} + f_{ik} f_{jl} + f_{il} f_{jk}). \end{aligned} \quad (17)$$

2.4. Linear superposition principle

From the previous discussion, we know the bilinear equation (13) possesses infinitely many exponential function solutions. We are interested in constructing some subset of such solutions which forms a nonzero linear space. The idea is to find the conditions such that for some given solutions, their linear combinations will still be solutions.

Suppose that $P(x)$ is a polynomial in $x \in \mathbb{R}^M$ (P may be not an even function). Assume that p is an integer no less than 2. Consider a generalized bilinear equation

$$P(D_p) f \cdot f = 0. \quad (18)$$

Let N be a positive integer. We define N -wave variables

$$\eta_m := \beta_{m,0} + \beta_{m,1} x_1 + \beta_{m,2} x_2 + \cdots + \beta_{m,M} x_M, \quad 1 \leq m \leq N \quad (19)$$

and exponential wave functions

$$f_m := e^{\eta_m}, \quad 1 \leq m \leq N, \quad (20)$$

where $\beta_{m,j}$, $1 \leq m \leq N$, $0 \leq j \leq M$, are all real constants.

We are interested in finding the conditions for a linear combination of functions

$$f = \varepsilon_1 f_1 + \varepsilon_2 f_2 + \cdots + \varepsilon_N f_N, \quad (21)$$

where ε_m , $1 \leq m \leq N$, are arbitrary real constants, and f_m , $1 \leq m \leq N$, are solutions of bilinear equation (18).

As in [4], we can compute that

$$\begin{aligned} P(D_p) f \cdot f &= \sum_{j,k=1}^N \varepsilon_j \varepsilon_k P(D_p) e^{\eta_j} \cdot e^{\eta_k} \\ &= \sum_{j,k=1}^N \varepsilon_j \varepsilon_k P(\beta_{j,1} + \alpha \beta_{k,1}, \dots, \beta_{j,M} + \alpha \beta_{k,M}) e^{\eta_j + \eta_k} \\ &= \sum_{j=1}^N \varepsilon_j^2 P(\beta_{j,1} + \alpha \beta_{j,1}, \dots, \beta_{j,M} + \alpha \beta_{j,M}) e^{2\eta_j} + \sum_{1 \leq j < k \leq N} \varepsilon_j \varepsilon_k e^{\eta_j + \eta_k} [P(\beta_{j,1} + \alpha \beta_{k,1}, \\ &\quad \dots, \beta_{j,M} + \alpha \beta_{k,M}) + P(\beta_{k,1} + \alpha \beta_{j,1}, \dots, \beta_{k,M} + \alpha \beta_{j,M})]. \end{aligned}$$

Because ε_j and ε_k , $1 \leq j, k \leq N$ are arbitrary constants, the function f , defined by (21), solves bilinear equation (18) if and only if for $1 \leq j \leq k \leq N$, it is true that

$$P(\beta_{j,1} + \alpha\beta_{k,1}, \dots, \beta_{j,M} + \alpha\beta_{k,M}) + P(\beta_{k,1} + \alpha\beta_{j,1}, \dots, \beta_{k,M} + \alpha\beta_{j,M}) = 0. \quad (22)$$

When $p = 2$, since $P(0) = 0$ and $P(-x) = P(x)$, thus (22) is reduced to

$$P(\beta_{j,1} - \beta_{k,1}, \dots, \beta_{j,M} - \beta_{k,M}) = 0, \quad (23)$$

for $1 \leq j < k \leq N$.

The above conditions are a collection of nonlinear algebraic equations, which paves a way of finding exact solutions by solving a group of algebraic equations rather than a group of nonlinear partial differential equations.

We now summarize our discussion as follows.

Basic fact: (Linear superposition principle [4]) Let $P(x)$ be a polynomial in $x \in \mathbb{R}^M$ and η_m , $1 \leq m \leq N$, be the N -wave variables $\eta_m = \sum_{j=1}^M \beta_{m,j} x_j$, $1 \leq m \leq N$, where $\beta_{m,j}$'s are all constants. Then any linear combination of the exponential waves $f_m = e^{\eta_m}$, $1 \leq m \leq N$, solves the bilinear differential equation (18) if and only if the conditions in (22) are satisfied.

Remark 1. In the above linear superposition principle if $\beta_{m,j}$'s are complex, then linear combinations provide a large class of complex valued function solutions.

By the proved linear superposition principle, we know for some bilinear partial differential equations under certain conditions, there exist infinite dimensional linear space consists of solutions although all the solutions do not form a linear space. Such a kind of examples can be seen in the works [9,10]. In subsets of the solutions to bilinear PDEs (18), we are particularly interested in that spanned by $f_m = e^{\eta_m}$ with

$$\eta_m = b_1 k_m^{\alpha_1} x_1 + b_2 k_m^{\alpha_2} x_2 + \dots + b_M k_m^{\alpha_M} x_M, \quad 1 \leq m \leq N, \quad (24)$$

where $b_j \in \mathbb{R}$, $\alpha_j \in \mathbb{Z}$, for $j = 1, \dots, M$, are fixed numbers, $k_m \in \mathbb{R}$, $m = 1, \dots, N$, are arbitrary.

Theorem 1. Let $P(x)$ be a polynomial in $x \in \mathbb{R}^M$ and η_m , $1 \leq m \leq N$, be the N -wave variables $\eta_m = \sum_{j=1}^M b_j k_m^{\alpha_j} x_j$, for $1 \leq m \leq N$, where b_j 's are all real constants. Then any linear combination of the exponential waves $f_m = e^{\eta_m}$, $1 \leq m \leq N$, solves the bilinear differential equation $P(D_p)f \cdot f = 0$ if and only if

$$P(b_1(k_j^{\alpha_1} + \alpha k_m^{\alpha_1}), \dots, b_M(k_j^{\alpha_M} + \alpha k_m^{\alpha_M})) + P(b_1(k_m^{\alpha_1} + \alpha k_j^{\alpha_1}), \dots, b_M(k_m^{\alpha_M} + \alpha k_j^{\alpha_M})) = 0, \quad (25)$$

for $1 \leq j \leq m \leq N$.

Moreover, (25) is true for arbitrary $k_1, k_2 \in \mathbb{R}$ if and only if it is also true for arbitrary constants $k_1, k_2 \in \mathbb{C}$.

Remark 2. Theorem 1 works also for complex k_m 's, which leads to complex-valued linear combination. However, we are interested only in real solutions. Is it possible for us to choose suitable coefficients to get real solutions?

From the second part of Theorem 1, we see that if (25) is true for arbitrary $k_1, k_2 \in \mathbb{R}$. Then N can be any positive integer and k_m , $1 \leq k_m \leq N$ are arbitrary complex numbers.

Let $N \in \mathbb{N}$, $i = \sqrt{-1}$. Suppose $k_m = \gamma_m + i\delta_m$, $\gamma_m, \delta_m, \theta_m \in \mathbb{R}$, $m = 1, 2, \dots, N$. Suppose that (25) is satisfied for any $k_1, k_2 \in \mathbb{R}$. Let the functions $\eta_m = \sum_{j=1}^M b_j k_m^{\alpha_j} x_j$, $f_m = e^{\eta_m + i\theta_m}$, for $m = 1, 2, \dots, N$. It is easy to check that the functions f_m and $\bar{f}_m$ are all solutions of the bilinear equation $P(D_p)f \cdot f = 0$.

By the superposition principle, we know

$$\sum_{m=1}^N \frac{\varepsilon_m}{2} (f_m + \bar{f}_m) = \sum_{m=1}^N \varepsilon_m e^{\text{Re}(\eta_m)} \cos(\text{Im}(\eta_m) + \theta_m) \quad (26)$$

is a real valued solution to (18) for arbitrary $\varepsilon_m \in \mathbb{R}$, $1 \leq m \leq N$. For some $m \in \{1, 2, \dots, N\}$, it may happen that $\delta_m = 0$. This implies that $k_m \in \mathbb{R}$ and η_m is a real wave function. Then we have a real solution of (18).

$$\varepsilon_m e^{\text{Re}(\eta_m)} \cos(\text{Im}(\eta_m) + \theta_m) = \varepsilon'_m e^{\eta_m}, \quad (27)$$

where $\varepsilon'_m = \varepsilon_m \cos(\theta_m)$ is a real constant.

In general, we have following result.

Theorem 2. Let $P(x)$ be a polynomial in $x \in \mathbb{R}^M$. Suppose that $N_1, N_2 \in \mathbb{N}$, $N = N_1 + N_2$, $b_j \in \mathbb{R}$, $\alpha_j \in \mathbb{Z}$ for $j = 1, \dots, M$, are all fixed. The N -wave variables $\eta_m = \sum_{j=1}^M b_j k_m^{\alpha_j} x_j$, $1 \leq m \leq N$ with parameters

$$\begin{cases} k_m \in \mathbb{R}, & 1 \leq m \leq N_1, \\ k_m = \gamma_m + i\delta_m, & \gamma_m, \delta_m \in \mathbb{R}, \delta_m \neq 0, N_1 < m \leq N. \end{cases} \quad (28)$$

If (25) is true for arbitrary $k_j, k_m \in \mathbb{R}, 1 \leq j, m \leq N$. Then any linear (real) combination of the waves f_m defined by

$$f_m = \begin{cases} e^{\eta_m}, & 1 \leq m \leq N_1, \\ e^{\operatorname{Re}(\eta_m)} \cos(\operatorname{Im}(\eta_m) + \theta_m), & N_1 < m \leq N, \end{cases} \quad (29)$$

with parameters $\theta_m, N_1 < m \leq N$ being arbitrary real constants, solves the bilinear differential equation $P(D_p)f \cdot f = 0$.

Remark 3. For any fixed $m, N_1 < m \leq N$, when we take $\theta_m = 0, -\pi/2$, respectively, we have the following two solutions in the vector space of solutions:

$$e^{\operatorname{Re}(\eta_m)} \cos(\operatorname{Im}(\eta_m)), \quad e^{\operatorname{Re}(\eta_m)} \sin(\operatorname{Im}(\eta_m)). \quad (30)$$

Any other linear combinations form the same space of solutions as the one spanned by the functions in (30).

In the case of $\operatorname{Re}(\eta_m) = 0, N_1 < m \leq N_2$. Applying the linear superposition principle, we will get linear combinations of trigonometric traveling waves which are solutions to the bilinear equation.

Remark 4. Suppose $p = 2, N_2 = 0$ and $N_1 = 2K$ is even. Assume that $\eta_{2j} = -\eta_{2j-1}, j = 1, \dots, K$ and conditions of Theorem 2 are satisfied. Then bilinear equation $P(D)f \cdot f = 0$ possesses solution

$$\sum_{j=1}^{N_1} \varepsilon_j e^{\eta_j} = \sum_{j=1}^K [(\varepsilon_{2j-1} + \varepsilon_{2j}) \cosh(\eta_{2j-1}) + (\varepsilon_{2j-1} - \varepsilon_{2j}) \sinh(\eta_{2j-1})]. \quad (31)$$

Obviously, this is a generalization of [11].

If we choose $\varepsilon_{2j} = \varepsilon_{2j-1}, j = 1, \dots, K$ then

$$\sum_{j=1}^{N_1} \varepsilon_j e^{\eta_j} = 2 \sum_{j=1}^K \varepsilon_{2j-1} \cosh(\eta_{2j-1}). \quad (32)$$

If we choose $\varepsilon_{2j} = -\varepsilon_{2j-1}, j = 1, \dots, K$ then

$$\sum_{j=1}^{N_1} \varepsilon_j e^{\eta_j} = 2 \sum_{j=1}^K \varepsilon_{2j-1} \sinh(\eta_{2j-1}). \quad (33)$$

3. Applications to bilinear equations

First, we consider Hirota bilinear equations.

Example 1. We consider the bilinear KP equation

$$(D_x^4 + D_x D_t + D_y^2)f \cdot f = 0. \quad (34)$$

For a fixed positive integer $N \geq 2$, consider

$$\eta_m = k_m x + b_1 k_m^2 y + b_2 k_m^3 t, \quad 1 \leq m \leq N. \quad (35)$$

Then by the linear superposition principle, we get

$$(k_j - k_m)[(k_j - k_m)^3 + b_2(k_j^3 - k_m^3)] + b_1^2(k_j^2 - k_m^2)^2 = 0, \quad 1 \leq j < m \leq N. \quad (36)$$

The parameters b_1 and b_2 should satisfy

$$\begin{cases} b_1^2 + b_2 + 1 = 0, \\ -b_2 - 4 = 0. \end{cases} \quad (37)$$

Therefore, we have solutions $b_1 = \sqrt{3}, b_2 = -4$ and $b_1 = -\sqrt{3}, b_2 = -4$.

Suppose that $N_1, N_2 \in \mathbb{N}, N = N_1 + N_2$. The N -wave variables $\eta_m = k_m x + b_1 k_m^2 y + b_2 k_m^3 t, 1 \leq m \leq N$ with parameters $k_m, 1 \leq m \leq N$, defined by (28). Moreover, for $N_1 < m \leq N$, we have

$$\begin{aligned} \eta_m &= k_m x + b_1 k_m^2 y + b_2 k_m^3 t \\ &= \gamma_m x + (\gamma_m^2 - \delta_m^2) b_1 y - 4(\gamma_m^3 - 3\gamma_m \delta_m^2) t + i[\delta_m x + 2\gamma_m \delta_m b_1 y - 4(3\gamma_m^2 \delta_m - \delta_m^3) t]. \end{aligned} \quad (38)$$

By [Theorem 2](#), any linear (real) combination of the exponential and trigonometric waves

$$f = \sum_{m=1}^{N_1} \varepsilon_m e^{k_m x + b_1 k_m^2 y - 4k_m^3 t} + \sum_{m=N_1+1}^N \varepsilon_m e^{\gamma_m x + (\gamma_m^2 - \delta_m^2) b_1 y - 4(\gamma_m^3 - 3\gamma_m \delta_m^2) t} \\ \times \cos(\delta_m x + 2\gamma_m \delta_m b_1 y - 4(3\gamma_m^2 \delta_m - \delta_m^3) t + \theta_m) \quad (39)$$

solves the bilinear KP equation with $b_1 = \sqrt{3}$ or $b_1 = -\sqrt{3}$.

Example 2. We introduce a polynomial

$$P(x, y, z, t) = c_1 x^4 + c_2 x^2 y + c_3 x z + c_4 x t + c_5 y^2 \quad (40)$$

and define weights of the independent variables:

$$(w(x), w(y), w(z), w(t)) = (1, 2, 3, 3). \quad (41)$$

The corresponding Hirota bilinear equation reads

$$P(D_x, D_y, D_z, D_t) f \cdot f = 2c_1 (ff_{4x} - 4f_x f_{3x} + 3f_{xx}^2) + 2c_3 (ff_{xz} - f_x f_z) \\ + 2c_4 (ff_{xt} - f_x f_t) + 2c_5 (ff_{yy} - f_y^2) \\ = (c_1 D_x^4 + c_3 D_x D_z + c_4 D_x D_t + c_5 D_y^2) f \cdot f = 0. \quad (42)$$

Let us take the wave variables

$$\eta_m = k_m x + b_1 k_m^2 y + b_2 k_m^3 z + b_3 k_m^3 t, \quad 1 \leq m \leq N. \quad (43)$$

Suppose that a linear subspace of N -wave solutions is given by

$$f = \sum_{m=1}^N \varepsilon_m f_m = \sum_{m=1}^N \varepsilon_m e^{\eta_m}. \quad (44)$$

Since the solutions of (42) do not depend on c_2 , taking $\tilde{P}(x, y, z, t) = c_1 x^4 + c_3 x z + c_4 x t + c_5 y^2$, applying [Theorem 1](#) for $\tilde{P}$, b_m , $m = 1, 2, 3$, need satisfy

$$\begin{cases} c_5 b_1^2 = 3c_1, \\ c_3 b_2 + c_4 b_3 = -4c_1. \end{cases} \quad (45)$$

We know if $c_1 c_5 > 0$, c_3, c_4 not all zero, then (45) has real solutions. Thus the corresponding equation (25) is satisfied. Suppose that k_m, η_m , $1 \leq m \leq N$ are defined by (28) and (29), respectively. Moreover, for $N_1 < m \leq N$, we have

$$\eta_m = k_m x + b_1 k_m^2 y + b_2 k_m^3 z + b_3 k_m^3 t \\ = \gamma_m x + (\gamma_m^2 - \delta_m^2) b_1 y + (\gamma_m^3 - 3\gamma_m \delta_m^2) (b_2 z + b_3 t) + i[\delta_m x + 2\gamma_m \delta_m b_1 y + (3\gamma_m^2 \delta_m - \delta_m^3) (b_2 z + b_3 t)]. \quad (46)$$

We can apply [Theorem 2](#). Any linear (real) combination of the exponential and trigonometric waves

$$f = \sum_{m=1}^{N_1} \varepsilon_m e^{\eta_m} + \sum_{m=N_1+1}^N \varepsilon_m e^{\operatorname{Re}(\eta_m)} \cos(\operatorname{Im}(\eta_m) + \theta_m) \quad (47)$$

solves the bilinear differential equation (42).

Note this example comes from [9] with $c_2 = 0$. We do not need extra conditions for the existence of complexitons. Since N_1 and N_2 are arbitrary, there exist a vector space consists of part of the solutions to the bilinear differential equation (42) with (uncountably) infinite dimension.

Example 3. Let a polynomial of weight 4 be given by

$$P = c_1 x^4 + c_2 x^2 y + c_3 x^2 z + c_4 x t + c_5 y^2 + c_6 y z + c_7 z^2, \quad (48)$$

with given weights

$$(w(x), w(y), w(z), w(t)) = (1, 2, 2, 3). \quad (49)$$

This example also comes from [9]. We are considering the bilinear equation

$$\begin{aligned} P(D_x, D_y, D_z, D_t)f \cdot f &= 2c_1(ff_{4x} - 4f_x f_{3x} + 3f_{xx}^2) + 2c_4(ff_{xt} - f_x f_t) + 2c_5(ff_{yy} - f_y^2) \\ &\quad + 2c_6(ff_{yz} - f_y f_z) + 2c_7(ff_{zz} - f_z^2) \\ &= 0. \end{aligned} \quad (50)$$

The solutions do not depend on c_2 and c_3 , the discussion in [9] was not correct. Let us define wave variables

$$\eta_m = k_m x + b_1 k_m^2 y + b_2 k_m^2 z + b_3 k_m^3 t, \quad 1 \leq m \leq N. \quad (51)$$

Suppose that the linear subspace of N -wave solutions is

$$f = \sum_{m=1}^N \varepsilon_m f_m = \sum_{m=1}^N \varepsilon_m e^{\eta_m}. \quad (52)$$

Based on Theorem 1, b_m , $m = 1, 2, 3$, need satisfy

$$\begin{cases} c_5 b_1^2 + c_6 b_1 b_2 + c_7 b_2^2 = 3c_1, \\ c_4 b_3 = -4c_1. \end{cases} \quad (53)$$

It is easy to see that $b_3 = -4c_1/c_4$ if $c_4 \neq 0$ or b_3 is arbitrary if $c_1 = c_4 = 0$. Denote $\Delta = c_6^2 - 4c_5 c_7$, if $\Delta > 0$ then (b_1, b_2) form a hyperbola; if $\Delta < 0$ and $c_1 c_5 > 0$ then (b_1, b_2) form an ellipse; in the case of $\Delta = 0$ and $c_1 c_5 > 0$ or $c_1 c_7 > 0$, (b_1, b_2) form two straight lines.

In general, there are several cases in which there are non-zero b_1 , b_2 and b_3 solving (53). Suppose that k_m , η_m , $1 \leq m \leq N$ are defined by (28) and (29), respectively. Moreover, for $N_1 < m \leq N$, we have

$$\begin{aligned} \eta_m &= k_{N+m} x + b_1 k_{N+m}^2 y + b_2 k_{N+m}^2 z + b_3 k_{N+m}^3 t \\ &= \gamma_m x + (\gamma_m^2 - \delta_m^2)(b_1 y + b_2 z) + (\gamma_m^3 - 3\gamma_m \delta_m^2) b_3 t + i[\delta_m x + 2\gamma_m \delta_m(b_1 y + b_2 z) + (3\gamma_m^2 \delta_m - \delta_m^3) b_3 t]. \end{aligned} \quad (54)$$

We can apply Theorem 2. Any linear (real) combination of the waves f_m , $1 \leq m \leq N$, defined by (29), solves the bilinear differential equation (50).

Example 4. In this example, we have weights $(w(x), w(y), w(z), w(t)) = (1, -1, -2, 3)$ with negative weight components. Then, consider a homogeneous polynomial of weight 2

$$P = c_1 t y + c_2 x^3 y + c_3 x^4 y^2 + c_4 x^4 z + c_5 z^2 t^2 + c_6 x^2. \quad (55)$$

This example is similar to but different from Example 5 in [9]. Let us take the wave variables

$$\eta_m = k_m x + b_1 k_m^{-1} y + b_2 k_m^{-2} z + b_3 k_m^3 t, \quad 1 \leq m \leq N. \quad (56)$$

The corresponding Hirota bilinear equation reads

$$\begin{aligned} P(D_x, D_y, D_z, D_t)f \cdot f &= 2c_1(ff_{ty} - f_t f_y) + 2c_2(ff_{3xy} - 3f_x f_{xy} + 3f_{xx} f_{xy} - f_{xxx} f_y) + 2c_3(ff_{4xyy} \\ &\quad - 4f_x f_{3xy} + 6f_{xx} f_{xyy} - 4f_{3x} f_{xy} + f_{4x} f_{yy} - 2f_y f_{4xy} + 8f_{xy} f_{3xy} - 6f_{xxy}^2) \\ &\quad + 2c_5(ff_{tzz} - 2f_z f_{tz} + f_{zz} f_t - 2f_t f_{zz} + 2f_{tz}^2) + 2c_6(ff_{xx} - f_x^2). \end{aligned} \quad (57)$$

It possesses an N -wave solution

$$f = \sum_{m=1}^N \varepsilon_m f_m = \sum_{m=1}^N \varepsilon_m e^{\eta_m}, \quad (58)$$

where ε_m 's and k_m 's are arbitrary, and b_j , $1 \leq j \leq 3$, satisfy

$$\begin{cases} 3c_2 b_1 = -c_6, \\ c_1 b_1 b_3 + c_2 b_1 = 0, \\ c_3 b_1^2 = c_5 (b_2 b_3)^2 = 0. \end{cases} \quad (59)$$

We have a solution $b_1 = -\frac{c_6}{3c_2}$, $b_3 = -\frac{c_2}{c_1}$ and b_2 is an arbitrary real number when $c_3 = c_5 = 0$ and $c_1 c_2 \neq 0$. (Note $c_4 D_x^4 D_z f \cdot f = 0$.) Eq. (57) reduces to

$$(c_1 D_t D_y + c_2 D_x^3 D_y + c_6 D_x^2) f \cdot f = 0. \quad (60)$$

Suppose that $k_m, \eta_m, 1 \leq m \leq N$, are defined by (28) and (29), respectively. Moreover, for $N_1 < m \leq N$ we compute

$$\begin{aligned}\eta_m &= k_m x + b_1 k_m^{-1} y + b_2 k_m^{-2} z + b_3 k_m^3 t \\ &= \gamma_m x + \frac{\gamma_m}{\sqrt{\gamma_m^2 + \delta_m^2}} b_1 y + \frac{\gamma_m^2 - \delta_m^2}{\gamma_m^2 + \delta_m^2} b_2 z + (\gamma_m^3 - 3\gamma_m \delta_m^2) b_3 t \\ &\quad + i[\delta_m x - \frac{\delta_m}{\sqrt{\gamma_m^2 + \delta_m^2}} b_1 y - \frac{2\gamma_m \delta_m}{\gamma_m^2 + \delta_m^2} b_2 z + (3\gamma_m^2 \delta_m - \delta_m^3) b_3 t].\end{aligned}\quad (61)$$

We use Theorem 2. Any linear (real) combination of waves $f_m, 1 \leq m \leq N$, defined by (29), solves the bilinear differential equation (60).

Now we consider two examples of generalized bilinear equations with the D_p -operators under $p = 3$.

Example 5. We consider a polynomial

$$P(x, y, z, t) = c_1 x^5 + c_2 x^3 y + c_3 x^2 z + c_4 x y^2 + c_5 x t + c_6 y z \quad (62)$$

with weights

$$(w(x), w(y), w(z), w(t)) = (1, 2, 3, 4). \quad (63)$$

This example is similar to but different from Example 1 in [4]. The corresponding Hirota bilinear equation is

$$\begin{aligned}P(D_{3,x}, D_{3,y}, D_{3,z}, D_{3,t})f \cdot f &= 2c_1(f_{5x}f - 5f_{4x}f_x + 10f_{3x}f_{2x}) + 6c_2f_{xx}f_{xy} + 2c_3f_{xxz}f + 2c_4f_{xyy} \\ &\quad + 2c_5(f_{xt}f - f_x f_t) + 2c_6(f_{yz}f - f_y f_z) = 0.\end{aligned}\quad (64)$$

Let us take the wave variables

$$\eta_m = k_m x + b_1 k_m^2 y + b_2 k_m^3 z + b_3 k_m^4 t, \quad 1 \leq j \leq N. \quad (65)$$

Suppose that the linear subspace of N -wave solutions is

$$f = \sum_{m=1}^N \varepsilon_m f_m = \sum_{m=1}^N \varepsilon_m e^{\eta_m}, \quad (66)$$

where $\varepsilon_m, 1 \leq m \leq N$, are arbitrary constants. The parameters b_1, b_2, b_3 satisfy

$$\begin{cases} c_6 b_1 b_2 - 3c_2 b_1 = 10c_1, \\ c_5 b_3 = -5c_1, \\ c_4 b_1^2 + c_5 b_1 b_2 + c_3 b_2 + c_5 b_3 = -c_1. \end{cases} \quad (67)$$

By symbolic computation, we get following result.

1. If $c_1 = 0$, then we have the solution

$$b_1 = 0, \quad b_2 = \begin{cases} 0, & \text{if } c_3 \neq 0, \\ \text{arbitrary}, & \text{if } c_3 = 0, \end{cases} \quad b_3 = \begin{cases} 0, & \text{if } c_5 \neq 0, \\ \text{arbitrary}, & \text{if } c_5 = 0. \end{cases} \quad (68)$$

Moreover, when $c_6 \neq 0$ and $\Delta = 9c_2^2 - 12c_2 c_3 c_4 / c_6 \geq 0$, we have another solution

$$b_1, \quad b_2 = \frac{3c_2}{c_6}, \quad b_3 = \begin{cases} 0, & \text{if } c_5 \neq 0, \\ \text{arbitrary}, & \text{if } c_5 = 0, \end{cases} \quad (69)$$

where b_1 is a real solution to the quadratic equation

$$c_4 b_1^2 + 3c_2 b_1 + 3c_2 c_3 / c_6 = 0. \quad (70)$$

2. If $c_1 c_4 c_5 c_6 \neq 0$, then we have the solution

$$b_2 = \frac{3c_2 b_1 + 10c_1}{c_6 b_1}, \quad b_3 = -\frac{5c_1}{c_5}, \quad (71)$$

where b_1 is a real solution to the equation

$$c_4 c_6 x^3 + 3c_2 c_6 x^2 + (3c_2 c_3 + 6c_1 c_6)x + 10c_1 c_3 = 0. \quad (72)$$

3. If $c_1 c_2 c_3 c_5 \neq 0$, and $c_6 = 0$, then we have the solution

$$b_1 = -\frac{10c_1}{3c_2}, \quad b_2 = \frac{4c_1}{c_3} - \frac{100c_1^2 c_4}{9c_2^2 c_3}, \quad b_3 = -\frac{5c_1}{c_5}. \quad (73)$$

Suppose that $k_m, \eta_m, 1 \leq m \leq N$, are defined by (28) and (29), respectively. Moreover, for $N_1 < m \leq N$ we have

$$\begin{aligned} \eta_m &= k_m x + b_1 k_{N+m}^2 y + b_2 k_m^3 z + b_3 k_m^4 t \\ &= \gamma_m x + (\gamma_m^2 - \delta_m^2) b_1 y + (\gamma_m^3 - 3\gamma_m \delta_m^2) b_2 z + (\gamma_m^4 + \delta_m^4 - 6\gamma_m^2 \delta_m^2) b_3 t \\ &\quad + i[\delta_m x + 2\gamma_m \delta_m b_1 y + (3\gamma_m^2 \delta_m - \delta_m^3) b_2 z + 4(\gamma_m^3 \delta_m - \gamma_m \delta_m^3) b_3 t]. \end{aligned} \quad (74)$$

We can apply Theorem 2. Any linear (real) combination of waves $f_m, 1 \leq m \leq N$, defined by (29), solves the bilinear differential equation (64).

Example 6. In this example, we have weights $(w(x), w(y), w(z), w(t)) = (1, -1, -2, 3)$ with negative weight components. Then, consider a homogeneous polynomial of weight 2

$$P = c_1 t y + c_2 x^3 y + c_3 x^4 y^2 + c_4 x^4 z + c_5 z^2 t^2 + c_6 x^2. \quad (75)$$

This polynomial P is the same as the one in Example 4. However, we consider the following generalized bilinear equation with $p = 3$:

$$\begin{aligned} P(D_{3,x}, D_{3,y}, D_{3,z}, D_{3,t}) f \cdot f &= 2c_1 (ff_{ty} - f_t f_y) + 6c_2 f_{xx} f_{xy} + 2c_3 (ff_{4xyy} + 4f_{3x} f_{xyy} + 6f_{xy}^2) \\ &\quad + 2c_4 (ff_{4xz} - 4f_x f_{3xz} + 6f_{xx} f_{xxz} + 4f_{3x} f_{xz} - f_{4x} f_z) \\ &\quad + 2c_5 (f_{tt} f_{zz} + 2f_{tz}^2) + 2c_6 (ff_{xx} - f_x^2). \end{aligned} \quad (76)$$

Let us take the wave variables

$$\eta_m = k_m x + b_1 k_m^{-1} y + b_2 k_m^{-2} z + b_3 k_m^3 t, \quad 1 \leq m \leq N. \quad (77)$$

By Theorem 2, when $c_4 = c_5 = 0$, the parameters b_1, b_2, b_3 need satisfy

$$\begin{cases} 24c_3 b_1^2 = 4c_6, \\ 8c_3 b_1^2 - 2c_1 b_1 b_3 = 0, \\ 2c_3 b_1^2 + 2c_1 b_1 b_3 + 6c_2 b_1 = -2c_6. \end{cases} \quad (78)$$

By using symbolic computation, we have the following solutions.

1. If $c_6 = 0$, then $b_1 = 0$ and b_2, b_3 are arbitrary real numbers.
2. If $c_1 c_2 \neq 0$ and $121c_3 c_6 - 54c_2^2 = 0$, we have a solution $b_1 = -\frac{11c_6}{18c_2}, b_3 = -\frac{22c_3 c_6}{9c_1 c_2}$ and b_2 is an arbitrary real number.

Suppose that $k_m, \eta_m, 1 \leq m \leq N$, are defined by (28) and (29), respectively. Moreover, for $N_1 < m \leq N$, we have

$$\begin{aligned} \eta_m &= k_m x + b_1 k_m^{-1} y + b_2 k_m^{-2} z + b_3 k_m^3 t \\ &= \gamma_m x + \frac{\gamma_m}{\sqrt{\gamma_m^2 + \delta_m^2}} b_1 y + \frac{\gamma_m^2 - \delta_m^2}{\gamma_m^2 + \delta_m^2} b_2 z + (\gamma_m^3 - 3\gamma_m \delta_m^2) b_3 t \\ &\quad + i[\delta_m x - \frac{\delta_m}{\sqrt{\gamma_m^2 + \delta_m^2}} b_1 y - \frac{2\gamma_m \delta_m}{\gamma_m^2 + \delta_m^2} b_2 z + (3\gamma_m^2 \delta_m - \delta_m^3) b_3 t]. \end{aligned} \quad (79)$$

Again we apply Theorem 2. Any linear (real) combination of waves $f_m, 1 \leq m \leq N$, defined by (29), solves the bilinear differential equation (76).

This example, together with Example 4, tells us that for $p = 2$ and $p = 3$, the formulation of mixed soliton and complexiton solutions is similar but b_1, b_2, b_3 should satisfy different algebraic equations. Actually, the corresponding bilinear equations are different, and they do represent different PDEs.

4. Concluding remarks

In this paper, we established the linear superposition principle to solitons and complexitons of both Hirota and generalized bilinear PDEs. The key is to take pairs of conjugate complex wave variables in formulating real solutions. By imposing coefficient conditions, we generated real subspaces of soliton and complexiton solutions to bilinear PDEs. We presented a few kinds of illustrative examples involving arbitrary coefficients to shed light on the existence of subspaces of complexitons to bilinear PDEs.

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